A Multi-Stage Real-Time CCTV Crime Detection Framework Using Spatial and Temporal Deep Learning Model

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***Abstract*— A major problem faced with CCTV-based security systems is monitoring as too much data is produced into a live view which people can hardly handle because at any given time only one stream can be monitored. The developments and applicability of deep learning-based automated detection of criminal activities show promising results. Unfortunately, the current approaches are mostly performed using a single-stage model, requiring full video processing for it to work as high computational-demand methods make it difficult to be real-time operations. This study proposes a multi-stage real-time crime detection system which enhances operational performance by premeditating video content while using its resource-demanding computational systems. This embedded motion-based filtering technique detects violent incidents through a light-weighted convolutional neural network and violence detection method with the help of a comprehensive deep learning model; hence, violence detection is done quite well. The 3D convolutional neural network is based real-time criminal detection system. The system also reduces unnecessary computing work since it uses its hierarchical structures to perform high accuracy in crime detection. The system presents reliable results in crime detection and works well on public surveillance data, making it a viable solution for real-time applications. The system is equipped with a live monitoring interface to produce alerts and prompt actions for human observers in demonstrating full surveillance capability.**

Keywords*— *CCTV Surveillance, Crime Detection, Deep Learning, Video Anomaly Detection, Violence Detection, Real-Time Monitoring, 3D Convolutional Neural Networks

I. Introduction (*Heading 1*)

"With smart cities, transportation hubs, educational institutions, and public spaces installed with Closed-Circuit Television (CCTV), there is an enormous flow of continuous video data. Monitoring such large flows of surveillance video data manually is inefficient, labour-intensive, and prone to human error. Therefore, in the computational vision and automatic security systems, automatic video-based crime and anomaly detection need to be undertaken seriously as major research areas."

Real-world anomaly detection in surveillance situations is genuinely challenging in the presence of varying backgrounds, lighting, crowd density, occlusions, and view. A heavy benchmark for anomaly detection in video surveillance was established in [1], stressing the need for weakly supervised deep learning methods that can characterize abnormal occurrences over long, untrimmed videos. The results demonstrate how deep neural networks can be used to learn discriminative representations of either raw or augmented video sequences. Greater momentum is given to the research on violence detection by way of structured datasets. The dataset (of large-scale real-world violence) introduced earlier [2] allows for deep learning to be thoroughly evaluated against elaborate surveillance setups. Earlier methods were based on motion estimation using optical flow analysis for detection of crowd aggression, while they were in progression in real time. Among these, Weitzer et al. exemplifies these efforts in [3]. The methods frequently worked pretty well in terms of computational efficiency, but they did not function well in terms of general reactivity and complicated scenarios.

With the introduction of deep convolutional neural networks, video action recognition has experienced a substantial shift. The distinguishing feature of [4] introduces an architecture with spatiotemporal convolutions that can learn both motion and appearance features concordantly at a much larger scale, arguably impacting datasets for action recognition. Consequently, it implicitly makes one thing very clear: in our deep architectures, it is time, for the first time, to directly model temporal dynamics. Additionally, an argument is made in [5] that the 3D convolutional neural network got a big shot in the arm in capturing spatial and temporal correlations while boosting performance on this recognition task. A different strategy for spatiotemporal information modeling is to employ the dual streams. The model of the dual stream of architecture was introduced with [6], understanding spatial information through RGB frames separate from temporal information extracted from optical flows that are associated with actions performed late after. This model, on its own, has become the base for several violence and anomaly detection techniques. Apart from development in architecture, the diversity of datasets forms an important key to generalizing ability. Automatic violence recognition stressing the importance of balanced and realistic data at the time of training deep learning models for surveillance [7].

While tremendous progress has been made in anomaly and violence detection, there are systems that work spatially and temporally in isolation. Real-time street view crime-detection calls for a multi-stage model that balances spatial feature extraction and modeling with that of a temporal sequence whilst maintaining computational efficiency. The system should incorporate both state-of-the-art deep spatial and deep learning models to allow real-time crime detection. That will ensure that the crime detection systems will be able to scale in massive terms and will also ensure pinpoint robustness.

II. Related Work

Featuring research efforts focused on direct surveillance intelligence, the models have also seen much interest in enhancing violence and anomaly detection models through the application of enhanced deep learning architectures. In [8], a succinct presentation of a framework was made that specifically catered to human violence recognition in surveillance systems, ideally being efficient with lightweight architectures for implementation in real time. The importance of keeping computational complexity while ensuring detection accuracy, especially in edge- and IoT-based CCTV systems, was also emphasized within this framework.

Also, throughout recent times, deep-learning–based anomaly detection technologies for surveillance video clips have acquired massive interest. A behind-the-scenes-all-extras study in [9] investigated deep neural network architectures for abnormal behavior detection in surveillance videos. It has enunciated the efficacy of convolutional feature extraction combined with temporal modus for improving recognition performance in the pursuit of modeling complex backgrounds. In their investigation on real-time violence detection within large public venues, such as stadiums, [10] deployed Bidirectional Long Short-Term Memory (Bi-LSTM) network models that employ deep learning methodologies to model temporal dependencies early detection of violent incidents in crowded environments. Estimated from humans' posed movements-and not from non-security-oriented appearance models-violence needs alternatives. Thus, one of them was outlined by [11], which used pose estimation and skeletal features are fields of identification for violence to be located in typical campus settings. That is how one systematically deflected the primary source of noise and illumination problems off of gathering joint movement data instead of raw pixel data.

An incomplete overview has been provided in [12] on the behavioral crowd, which talks about a taxonomy on deep learning approaches for the detection of anomalies; for the recognition of crowd emotion's behavior analysis. He went on to argue that actual scenario-based frameworks are essential to deal with dense crowds and dynamic considerations. Moreover, outside of surveillance, Deep Learning methods were used in useful and intelligent systems. In open waters with no paths, Deep Learning assisted navigation system was using spatial models to increase safety-based applications [13]. Even though the subtype model was developed to assist the visually challenged, this sort of aptness will be of paramount importance for real-time environmental understanding in safety-critical systems. Finding violence in multimedia content: [14] says that one could use deep learning to find violence in videos. Convolutional architectures could capture high-level semantic patterns with respect to activities in violent scenes and then maybe applied to surveillance networks or crime detection networks. Hybrid deep feature extraction with traditional machine learning classifiers was also utilized. [15] combined 3D convolutional neural networks with Support Vector Machines (SVM) for violence detection, showing that working with deep features can truly enhance decision-making with good classical classifiers.

Recently, there has been an increased effort in developing systems for intelligent classification examining alternative neural paradigms of pattern recognition. The weighted neural-network-based multi-class fault detection model proposed in [16] draws attention to the results of binary pattern encoding utilized for efficient classification. Exampled applied in the field of industrial fault diagnosis, the above approach, however, displays a huge potential for staged classification strategies in intricate detection systems. Real-time anomaly detection using the spatial dimensions from CNN features combined with bidirectional LSTM networks is examined in [17]. The spatial information was extracted from the convolutional network architecture, while the temporal dependencies were modeled using recurrent networks. The model that integrates CNN and Bi-LSM founded accurate abnormal-event recognition in surveillance networks, which emphasizes how important it is to harmonize the given spatial and temporal learning mechanisms Existing violence and anomaly detection have come a long way, but all methods still long only for single-stage architectures or seem to concentrate only on spatial or temporal modeling. A comprehensive multi-stage framework combining spatial feature extraction, temporal sequence modeling, and real-time optimization is required for the advancement of reliable and scalable crime detection systems using CCTV. The proposed framework fills this gap by incorporating spatial and temporal deep learning models in a structured multi-stage pipeline.

III. Proposed Methodology

A neural network crime detection system primarily using CCTV video data needs a method called the proposed framework, which applies a two-stage aparative deep learning model method for the detection by location and time for increased efficiency in accuracy. The first stage comprises a ConvNet to segregate those spatial features that make video patterns doubtful by stretching the individual video frames. The second stage, a temporal modeling network parses sequential frame representations of motion dynamics and the development of activities. The spatial and temporal decisions, derived from a decision fusion method, merge and minimize false alarms through accumulation. The operational program is designed to be time-effective, meta-consistent, and accurate for forensic surveillance deployment.

A. System Workflow

In the proposed architecture, multiple stages of processing are performed to analyze suspicious activities within surveillance camera systems. Instead of directly applying computationally intensive models to the whole video sequence, the video stream is progressively filtered by several stages. Each stage carries out a specific task and passes only the relevant part ahead to the next, and this is intended to minimize needless computation while achieving a high recall of detection. There are five stages in total following the processing pipeline. The first stage is responsible for motion-based filtering to trace unusual events within the video stream. The system uses a lightweight convolutional neural network to let each frame be processed in order to analyze individual frames for anomalous behaviors in the second stage. Here, the third stage then inspects suspicious segments for violent interactions. Following another level, primary events get acknowledged using a 3D convolutional neural network in relation to time in Stage-3. The fourth stage detects variance by setting alerts for quick decisions; it is interactive enough to ascertain the validity of an event with instantaneous action. It guarantees the surveillance for an environment that is both efficient and real-time in being processed. A table 1 in the paper summarizes the datasets for the respective stages in the proposed system.

TABLE I. DATASET DESCRIPTION

Dataset	Purpose	Classes	Type
RWF-2000	Violence Detection	Fight/non-fight	Video Clips
UCF-Crime	Temporal Crime Detection	Normal/Crime	Videos
XD-Violence	Anomaly Detection	Normal/Violence	Frames

B. *Stage-0: Motion Detection*

Phase-0 is the preliminary filter for the proposed system. This is the case when a considerable portion of the image sequences under real-world surveillance presents still views. Deep learning models working on such image frames cause unnecessary computational overhead. This issue is tackled by detecting motion to identify active periods within the video stream. Motion is evaluated with frame differences over several preceding frames. Any motion values above a threshold condition become candidate frames to retain for deep learning model calibration. This drastically diminishes the total number of frames that need to be processed by latter deep learning models, leading to a more efficient overall system.

C. *Stage-1: Abnormal Activity Detection*

Stage 1 detects abnormal activities on a frame level by utilizing convolutional neural networks described by MobileNetV2, at the same time extracting spatial features from individual frames, and thereby classifying those for their respective normal and abnormal activities. Its lightweight design ensures fast processing without compromising on the detection accuracy. Everything-but -abnormal frames is dropped, due to which any frames containing routine and regular activities like walking or sitting are not processed any further. The reason for incorporating this extra layer is to avoid falsely processing the footage of walking etc. so in stages where heavier and more intensive operations are only requisite.

D. *Stage-2: Violence Detection*

Now that we have examples of Stage-1 abnormality here, the analysis in Stage-2 will focus on detecting violence. Since the norm has been defined at the frame level, those dynamics/interactions occurring over multiple frames are irrelevant in this context; therefore, we need to look out for aggressive, violent behaviour in the temporal domain. The spatial features in this stage also continue to be extracted with a convolutional neural network backbone running on the basis of two consecutive frames. Only throughout this are the said features treated across different frames to exploit a very short term temporal informativity streaking in the video segment. Based on the motion interaction and pattern of each segment, the model provides either a label of fight or nonfight. This cycle is supposed to remove normal but abnormal activities and direct anything only marginally suspicious for more rigorous processing in the next checkpoint.

E. *Stage-3: Temporal Crime Detection*

The 5th layer, pretty much as with any other, has the main advantage of granting one the results context (since many frames are being considered) so that one can infer some semblance of rationality about the environment within which the event took place. It runs the video clips of a given duration and learns the temporal patterns of those activities related to the crime. It does this by identifying motion continuity and the growth of the activity through time, hence isolating normal activities of crime from a given time. Such discriminations are necessary for determining the proposed system, and further suggest a substantial increase in systematic understanding for the constellarium.

F. *Stage-4: Alert Generation and Operator Verification*

Once computer-aided crime detection takes place, it will process alerts, with the probability of criminal offense above the predefined threshold. To register the event, if crime probability exceeds the threshold, alerts are sent with timestamps and the corresponding confidence value. So, the real-time monitor interface will allow operators to see all such detected events and mark them as true while dealing with false positives. This may be taken to be its biggest characteristic as it enables the actual system implementation by putting back in human verification in the loop of detection of criminal activities. This in turn aims at minimizing the ill impact of false positives on the working of the system-especially with a real advance of environmental surveillance systems. Figure 1 describes the overview of the proposed multi stage CCTV crime detection system

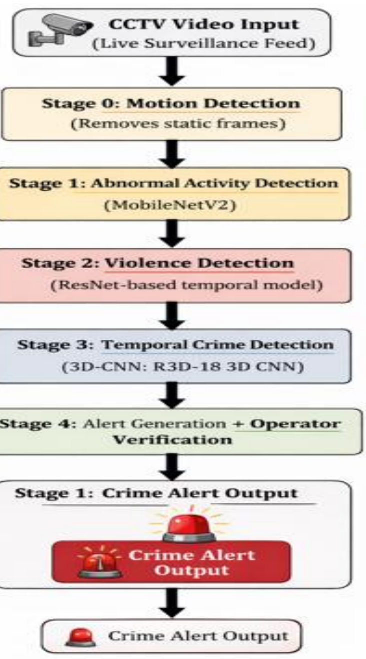

Fig. 1. Overview Multi-stage CCTV crime detection system

IV. EXPERIMENTAL SETUP

A. *Hardware and Implementation Details*

The researchers conducted all tests of their system in a personal computer environment to assess its ability to operate in real-time with restricted processing capabilities. The system was developed through Python programming which used the PyTorch deep learning framework as its foundation. The system used OpenCV to handle video processing and frame extraction tasks. The proposed system proved its operational capacity through training and inference tests which ran on CPU-based systems without needing advanced GPU equipment.

B. Datasets

The testing of the suggested system underwent evaluation through experiments conducted on publicly available surveillance footage and violence detection datasets. The Stage-2 model training for violence detection used video samples which contained both fighting and non-fighting activities. The Stage-3 model training process evaluated surveillance videos which displayed both typical human behavior and illegal activities. The datasets provide real-world surveillance scenarios which show different lighting conditions and various camera angles and multiple scene complexities to test the effectiveness of actual crime detection systems. The researchers used fixed-length video clips from the datasets to create temporal input which they used for model training in all stages through the division of the datasets into training and validation sets.

C. Training Configuration

The deep learning models used video clips which contained 16 consecutive frames as their input for temporal analysis. The training process required the frames to be resized into specific spatial dimensions. The model convergence process received enhancement through the implementation of standard normalization techniques. The models learned through binary classification objectives which used adaptive optimization techniques for their training process. The training process used early stopping together with learning rate scheduling to maintain stable performance while preventing overfitting.

D. Evaluation Metrics

The performance of the proposed system was evaluated using standard classification metrics including accuracy, precision, recall, and F1-score. Confusion matrices were used to analyze classification behavior at different stages of the pipeline. Since surveillance systems prioritize minimizing missed crime events, recall was considered an important evaluation metric during analysis. In addition, qualitative evaluation was performed through real-time testing using live camera input to validate the practical usability of the system.

V. RESULT AND DISCUSSION

A. Stage-1: Abnormal Activity Detection

The system deployed Stage-1 to perform initial surveillance frame filtering through a lightweight convolutional neural network. The stage aims to keep abnormal activities while decreasing the amount of normal footage which needs processing in subsequent stages. The experimental results demonstrate that the model successfully detects abnormal activities because it maintains all suspicious activities during its initial screening process. The system achieved its first stage performance evaluation by filtering 41% of normal frames which led to decreased processing requirements in upcoming evaluation phases. The high recall achieved at this stage confirms its effectiveness as an initial screening mechanism in the proposed system.

The system needs to achieve maximum recall because Stage-1 functions as an essential initial screening process. The current performance achieves acceptable results because the system can fix false positives which occur in subsequent processes. The system design guarantees that critical abnormal events will not be missed during the initial screening process. The system needs Stage-1 to handle highly sensitive details instead of achieving exact results through classification.

Figure 2 illustrate Stage-1 confusion matrix exceptional performance because it detects most abnormal activities through its primary filtering process. The system design uses misclassification of normal frames as abnormal to achieve its main function which protects against active crime detection failures.

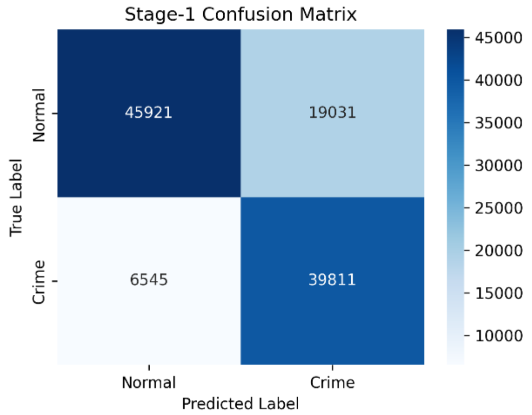

Fig. 2. Confusion matrix for Stage-1 Abnormal Activity detection

B. Stage-2: Violence Detection

Stage-2 examines questionable video clips to identify acts of violence. The evaluation results show that the confusion matrix demonstrates the model's ability to detect fights through high recall performance which enables it to identify most violent incidents. The system is considered to perform adequately because it incorrectly classifies non-violent actions as fights, leading to an increase in security alerts while actual violent events are often overlooked. The results show that Stage-2 successfully reduces abnormal behavior patterns to violence-related incidents which it uses to conduct time-based criminal investigations. Figure 3 demonstrate Stage-2 confusion matrix in strong fight detection recall which effectively detects most violent encounters. The system accepts this fight classification error because surveillance systems need to detect all violent events which customers want to track

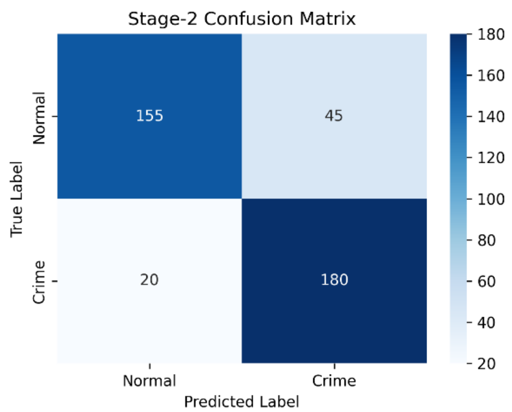

Fig. 3. Confusion matrix for Stage-2 violence detection.

C. Stage-3: Temporal Crime Detection

Stage-3 examines questionable video clips to identify acts of violence. The evaluation results show that the confusion matrix demonstrates the model's ability to detect fights through high recall performance which enables it to identify most violent incidents. The system is considered to perform adequately because it incorrectly classifies non-violent actions as fights, leading to an increase in security alerts while actual violent events are often overlooked. The results show that Stage-2 successfully reduces abnormal behavior patterns to violence-related incidents which it uses to conduct time-based criminal investigations. Figure 4 demonstrate Stage-3 confusion matrix where successful time-based crime pattern detection through its video sequence analysis. The system detects more crimes than usual but the system achieves its results because it understands contextual information better than frame-by-frame assessment.

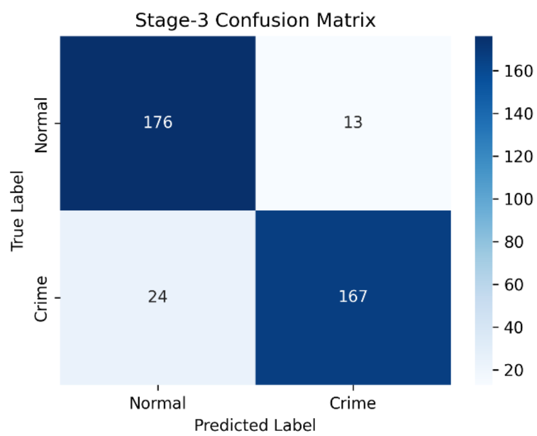

Fig. 4. Confusion matrix for Stage-3 Temporal Crime Detection.

The Table II presents evaluation results from all system stages which show that the initial system stages concentrate on recall because they aim to prevent event detection failures. The system uses a hierarchical structure that improves prediction accuracy while delivering results in real time with restricted processing power.

TABLE II. STAGE-WISE PERFORMANCE OF THE PROPOSED SYSTEM

Stage	Task	Accuracy	Precision	Recall	F1-score
Stage-1	Abnormal Detection	0.77	0.68	0.86	0.76
Stage-2	Violence Detection	0.84	0.80	0.90	0.85
Stage-3	Crime Detection	0.90	0.93	0.87	0.90

D. Discussion

The proposed system demonstrates its operational performance through stage-by-stage evaluation of its hierarchical design. The initial stages have been structured to achieve maximum recall which enables them to preserve all events that show potential links to suspicious activities or criminal behavior during their first assessment. The subsequent stages use temporal data to build better activity understanding through multiple frames of information. The multi-stage system prevents extra processing work because it stops normal behavior at initial check points while keeping essential parts for detailed review. The system maintains moderate accuracy because its design focuses on achieving high recall which helps detect all actual criminal activities needed for operational security in real-world surveillance systems. The experimental results show that the proposed multi-stage approach provides a practical balance between detection performance and real-time feasibility which makes it suitable for deployment in resource-constrained surveillance environments.

VI. CONCLUSION AND FUTURE WORK

This paper introduced a multi-stage, real-time CCTV-based crime detection framework aimed at improving the efficiency of surveillance processes. A three-tier system is employed that merges motion-based filtering with the detection of unusual activities, violence, and the recognition of crimes over time. The system achieves high crime detection accuracy because it uses multiple stages to filter video data which results in decreased computation needs. The experimental results show that the system achieves optimal detection results because the stage-wise method achieves balanced detection performance with energy-efficient operation which enables real-time usage on devices with restricted processing power. The alert generation module with operator verification system enhances usability because it allows operators to confirm which events have been detected. Future upgrades to the proposed system will involve the adoption of advanced temporal modeling strategies, aiming to strengthen its capacity to interpret complex actions in their respective contexts. The system will achieve better crime classification results through the integration of object detection modules which will identify weapons and suspicious objects. The system can extend its capabilities to enable monitoring of multiple cameras from a central location. The system will gain better real-world deployment performance through optimization methods which combine hardware acceleration with model compression techniques.

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